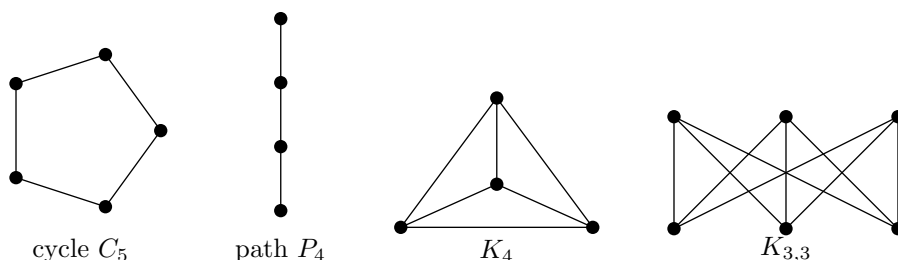


Chapter 1: Graphs, Subgraphs, Degrees, and Connections

1.1 Graphs and Subgraphs

A **graph** is a set of vertices (singular vertex) and edges, where every edge joins two vertices. A **simple** graph is one without loops or multiple edges; we will assume our graph is simple unless otherwise stated. Two vertices are said to be **neighbors** or **adjacent** if they are joined by an edge. A graph is **connected** if one can get from each vertex to every other vertex by a sequence of edges. The **components** of a graph are the connected pieces.

There are four common graphs. We use P_n to denote the **path** on n vertices, C_n to denote the **cycle** on n vertices, and K_n to denote the **complete graph** on n vertices. We also use $K_{r,s}$ to denote the **complete bipartite graph** constructed by taking a set of r vertices and another set of s vertices, and adding all possible edges between the two sets. The graph P_n has $n - 1$ edges, the graph C_n has n edges, the complete graph K_n has $\binom{n}{2}$ edges, and the complete bipartite graph $K_{r,s}$ has rs edges. Here are some pictures.

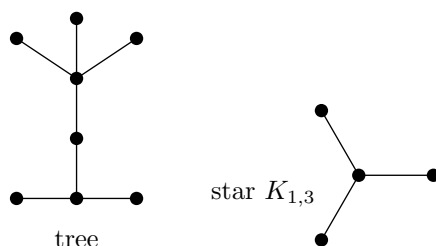


A **subgraph** of a graph has some of the vertices and some of the edges. An **induced subgraph** is one which contains all possible edges for its vertices. A **clique** is a set of vertices such that every pair of vertices are adjacent; equivalently, a subgraph that is complete. An **independent set** is a set of vertices such that no pair of vertices is adjacent.

EXAMPLE. The maximum size of an independent set is: $\lceil n/2 \rceil$ for P_n , $\lfloor n/2 \rfloor$ for C_n , 1 for K_n , and $\max\{r, s\}$ for $K_{r,s}$. The maximum size of a clique is 2

for P_n (unless $n = 1$), for C_n (unless $n = 3$) and for $K_{r,s}$.

A graph without a cycle is called **acyclic** or a **forest**. A **tree** is a connected acyclic graph. Here is a forest of two trees.



1.2 Degrees, Paths, and Eulerian Tours

The **degree** of a vertex v , denoted $d(v)$, is the number of edges incident with it (except loops count twice). A graph is **k -regular** if every vertex has degree k .

FIRST THEOREM OF GRAPH THEORY. *The sum of the degrees of the graph is twice the number of edges.*

PROOF. We use double counting. Let A be the number of ends of edges. Then A is twice the number of edges, since every edge has two ends. On the other hand, every end of an edge contributes one to the degree of that vertex. So summing up the degrees gives the total number of ends of edges. $\square$

It follows that in any graph the number of vertices of odd degree is even.

A **walk** is a sequence of edges. A **trail** is a walk without repeated edges (but can have repeated vertices). A **path** is a trail without repeated vertices. A **cycle** is almost a path except the first and last vertices are the same. A walk is **closed** if the first and last vertices are the same. The **length** of a walk is the number of edges.

There are several interesting properties of trees:

PROPERTIES OF TREES. *For every tree:*
 (a) *The number of edges is one less than the order.*

- (b) Every pair of vertices are joined by a unique path.
- (c) The removal of any edge disconnects the tree.
- (d) The addition of any edge creates a unique cycle.
- (e) If the tree has at least one edge, then it has at least two vertices of degree 1 (called **leaves**).

Indeed, one can characterize trees by such properties. It follows that:

OBSERVATION. *If a graph is connected, then the number of edges is at least one less than the order.*

A graph is **bipartite** if the vertices can be partitioned into two sets (called **partite sets**) such that every edge has an endpoint in both sets. Equivalently, one can color the vertices with two colors, say red and blue, such that both the red vertices and the blue vertices form an independent set. For example, trees and even cycles are bipartite. It can be shown that if a bipartite graph is regular, then the partite sets have equal size.

CHARACTERIZATION OF BIPARTITE GRAPHS. *A graph is bipartite if and only if it contains no odd cycle.*

PROOF. It is straightforward that an odd cycle is not bipartite. Thus every graph containing an odd cycle is not bipartite. To show the other direction, we provide a recipe to color the graph: either the recipe produces a bipartite coloring or it produces an odd cycle. Pick a vertex v and color in rounds: in round 0 color v red, in round 1 color its neighbors blue, in round 2 color their neighbors red, and so on. Note that two vertices that are adjacent are colored in the same round or in consecutive rounds. Suppose at the end there is an edge whose ends x and y have the same color. Then the two vertices were colored in the same round. For each vertex, we can construct a path from it back to v . Suppose the two paths back from x and y first meet up at vertex v' . Then the portion of the paths up to v' together with the edge xy form an odd cycle. $\square$

An **Eulerian tour** is a closed trail that uses every edge. A graph is **Eulerian** if it has an Eulerian tour.

CHARACTERIZATION OF EULERIAN GRAPHS. *A graph without isolated vertices has an Eulerian tour if and only if it is connected and every vertex has even degree.*

PROOF. (1) Assume we have an Eulerian tour. Every vertex other than the start must have even degree, since every time we visit a vertex we use up two edges incident with it. Similarly, the start vertex has even degree: apart from the first and last edge of the tour we use up the edges in pairs. It follows that every degree is even.

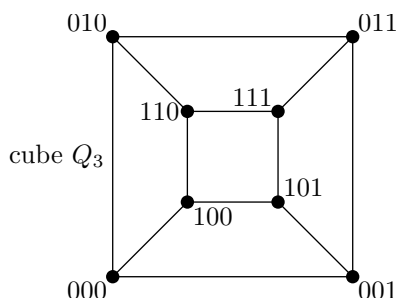
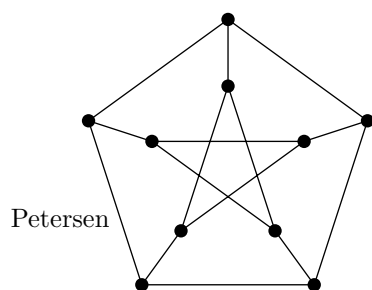
(2) Define a ***tour decomposition*** as a collection of tours that use up all the edges of the graph. Assume the graph has every vertex of even degree. Then we claim there is a tour decomposition.

One can construct a tour decomposition in a simple fashion: start tracing out a tour until one gets stuck. The only place one can get stuck is back where one started, since for every other vertex there is always another edge to leave it. Since a tour uses an even degree at each vertex, once we remove the first tour, what remains must have even degree throughout. Thus, one can repeat the process, thereby using up all the edges.

But what we want is an Eulerian tour. So take the tour decomposition $\mathcal{T}$ with the minimum number of tours. And suppose there is more than one tour in $\mathcal{T}$.

Then, because the original graph is connected, there must be somewhere two tours that share a vertex, call it p . We can re-organize things such that p is the start and finish of both tours. But then it is easy to merge the two tours, by continuing on the second after finishing the first. Hence we get a tour decomposition with fewer tours, a contradiction. That is, in fact, $\mathcal{T}$ contains an Eulerian tour. $\square$

We finish this section by introducing two common graphs. The Petersen graph has 10 vertices, is 3-regular, and the shortest cycle has length 5. The k -cube or hypercube Q_k is the graph whose vertices are all k -tuples with 0 and 1, and edges if the two k -tuples differ in exactly one position. It is bipartite, has 2^k vertices and is k -regular. The hypercube also has a recursive definition, being built up from smaller hypercubes. Here are the pictures.



1.3 Distance, Degrees, and Decompositions

The **degree sequence** of a graph is the sorted list of the degrees of its vertices. Any degree sequence is possible provided the sum of degrees is even. However, if one needs a simple graph, then there are several characterizations, the most famous of which is by Havel–Hakimi.

A **decomposition** of a graph is a partition of the edges (each edge is in exactly one part of the partition). The concept of Eulerian tour can be generalized to characterize graphs that can be decomposed into k trails.

Two graphs are **isomorphic** if there is a bijection between the vertex sets that preserves the edges. The **complement** of a simple graph has the same vertex set but the missing edges. A graph is **self-complementary** if it is isomorphic to its complement (e.g. P_4 or C_5). An exercise is to show that the order must be congruent to 0 or 1 modulo 4.

The **distance** from u to v is the length of the shortest path from u to v . The **eccentricity** of a vertex is the maximum distance of a vertex from it. The **radius** is the minimum eccentricity and the **center** is those vertices of minimum eccentricity. The **diameter** is the maximum eccentricity, or equivalently, the maximum distance between vertices. The **girth** is the shortest cycle length.

DISTANCE BOUNDS. *For any connected graph,*

(a) $\text{rad} \leq \text{diam} \leq 2 \text{rad}$.

(b) *If the graph is not a tree then* $\text{girth} \leq 2 \text{rad} + 1$

PROOF. (a) Consider any two vertices x and y , and let v be a vertex in the center (that is, having minimum eccentricity). Then there is a walk from x to y

via v ; thus the distance from x to y is at most 2rad .

(b) Suppose v is a vertex in the center. Consider a tree subgraph T formed by taking for each vertex a shortest path from it to v . Since the whole graph is not a tree, there is some edge not in T ; that together with T creates a cycle of length at most $2\text{rad}+1$. $\square$

The **Wiener index** of a graph is the sum of the distances between every pair of vertices. (If divide by $\binom{n}{2}$ then get the **average distance**.) For trees of order n , the Wiener index is maximum for a path and minimum for a star.